



London Ambulance Service



NHS Trust

Anatomy and Physiology – Part Component 2

Lymphatic System



Lymphatic System - Part Component 2

Objective

- Explain the structure and functions of the lymphatic system
- Understand which other systems it is linked to

Lymphatic System - Part Component 2

Functions:

Tissue Drainage:

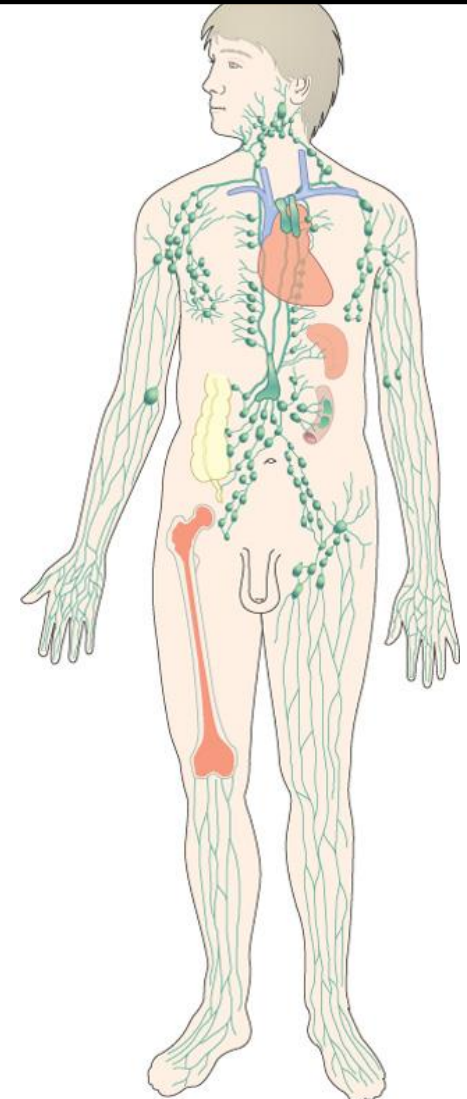
- 21L fluid from blood plasma is leaked per day in to tissues.
- Lymphatic system drains 3-4L of this from the tissues back in to circulation.

Absorption of fluid in small intestine

- Fat and fat soluble materials are absorbed into lymphatic vessels.

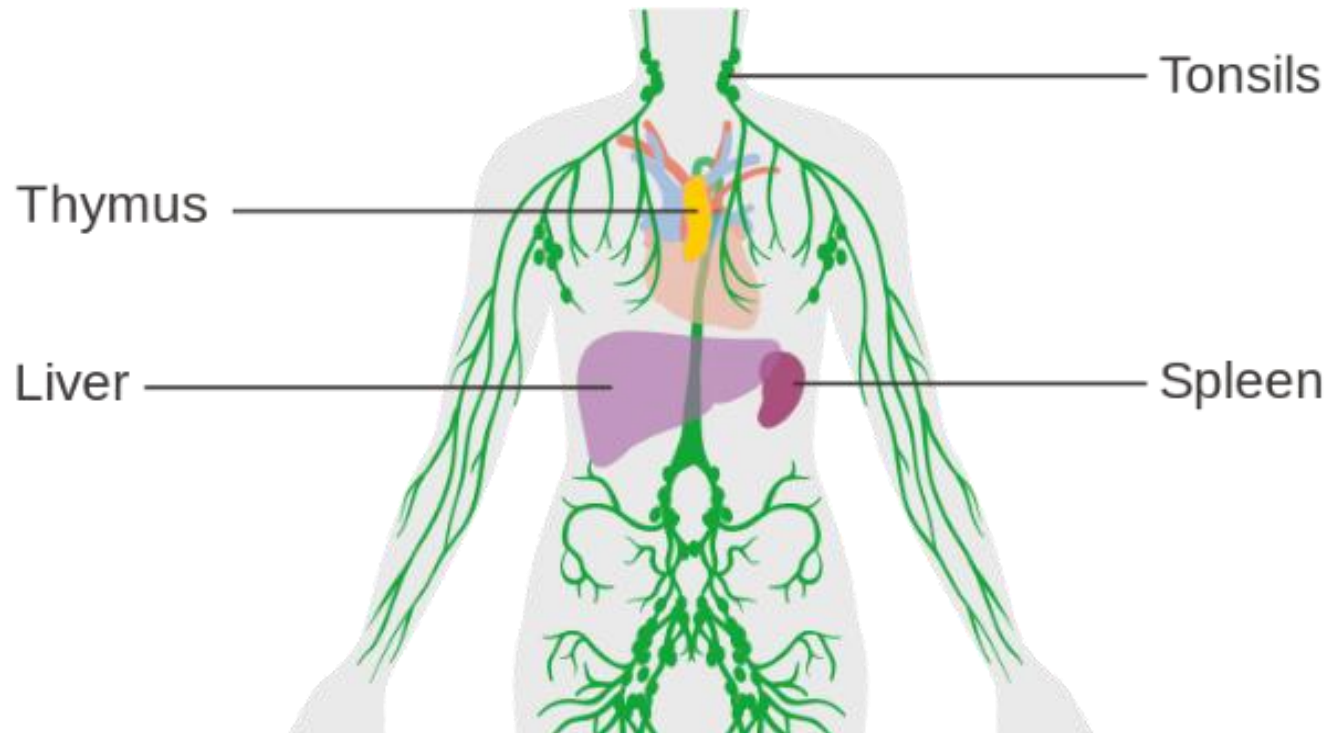
Immunity

- Produce and mature lymphocytes (white blood cells that are responsible for immunity).



Lymphatic System - Part Component 2

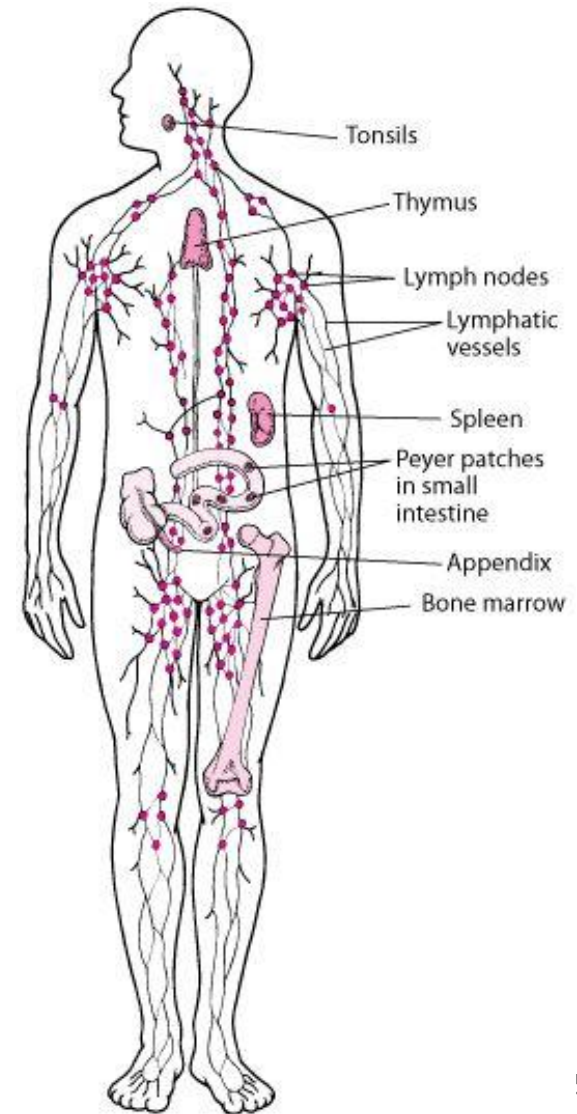
Main organs



Lymphatic System - Part Component 2

Consists of the following structures:

- Lymph
- Lymph vessels
- Lymph nodes
- Lymph organs (e.g. spleen and thymus)
- Diffuse lymphoid tissue (e.g. tonsils)
- Bone marrow





Lymphatic System - Part Component 2

Lymph:

- Clear watery fluid similar in composition to blood plasma but without the proteins.
- Transports plasma proteins that have leaked from capillaries back into the bloodstream.
- Transports large particles (e.g. bacteria and cell debris from damaged tissues) to be filtered and destroyed by lymph nodes
- Contains lymphocytes – originates from red bone marrow

Lymphatic System - Part Component 2

Lymph vessels:

Lymph capillaries

- Originate as blind-end tubes in interstitial (tissue) space
- Permeable to all interstitial fluid constituents (proteins and cell debris)

Large lymph vessels

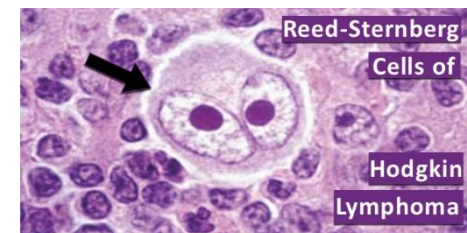
- Found alongside arteries and veins
- Contain valves for one way fluid flow – muscle in lymph vessels contract rhythmically to allow movement of fluid
- Lymph vessels become larger and eventually join to form two ducts which empty lymph into subclavian veins (veins in the neck)



Lymphatic System - Part Component 2

Common Pathology

- Lymphadenopathy
- Lymphoedema
- Lymphomas



Lymphatic System - Part Component 2

Questions?





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Anatomy and Physiology – Part Component 2

Urinary System



Urinary System - Part Component 2

Objective

Explain the structure and functions of the Urinary system



Urinary System - Part Component 2

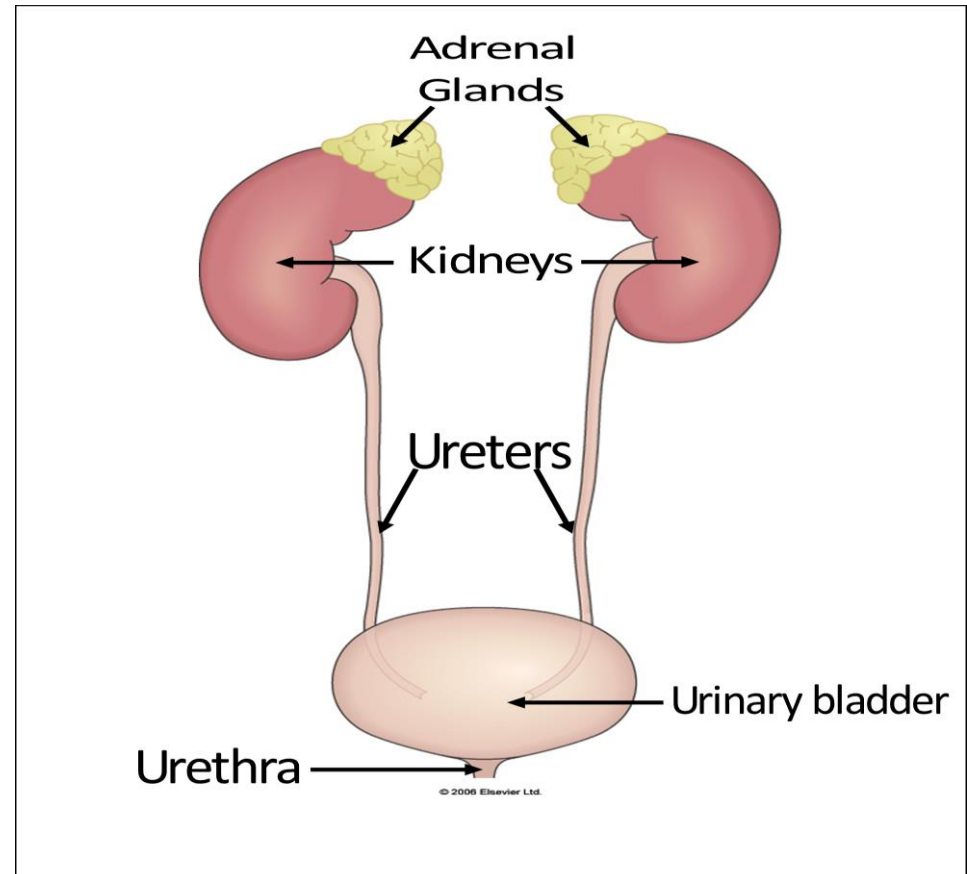
Functions

- Formation of urine
- Maintaining water, electrolyte and acid-base balance
- Excretion of waste products
- Regulation of red blood cell production
- Regulation of blood pressure

Urinary System - Part Component 2

Anatomy of the Urinary System

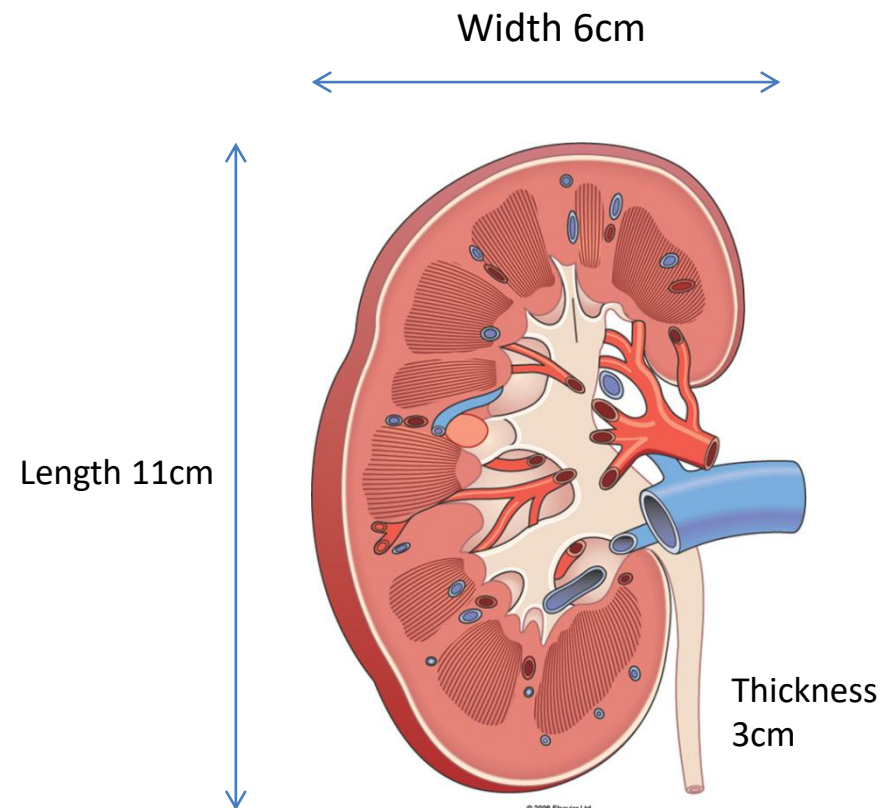
2 Kidneys
2 Ureters
1 Urinary Bladder
1 Urethra



Urinary System - Part Component 2

The Kidneys

- Bean-shaped organ
- Weight 150g



Urinary System - Part Component 2

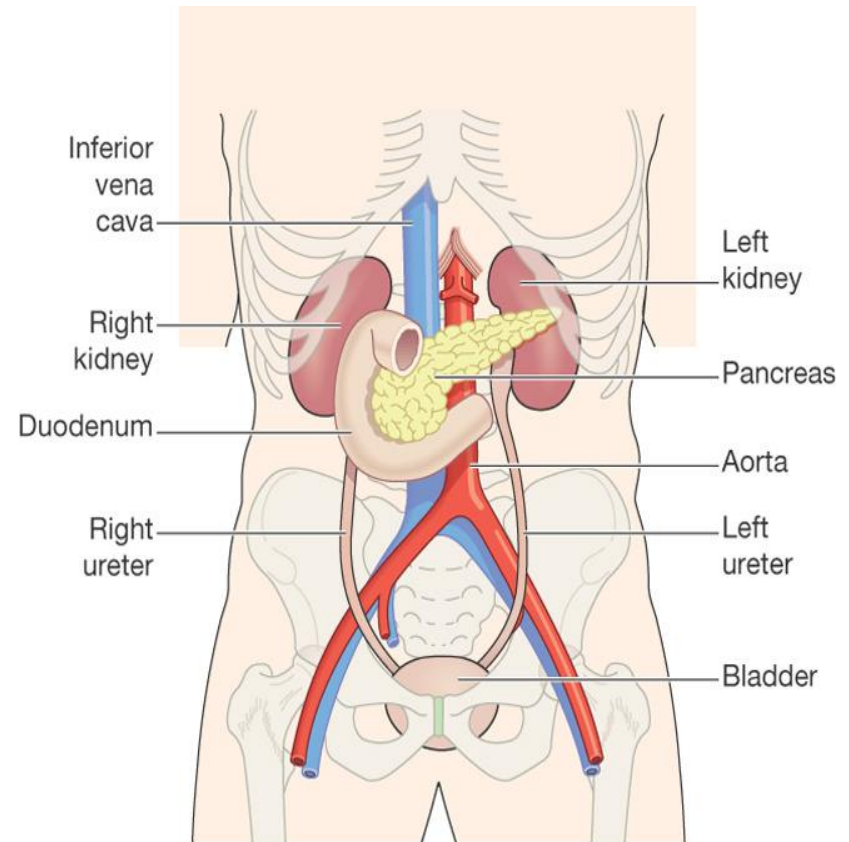
The Kidneys Position

The Kidneys sit in the abdominal cavity

The 2 kidneys sit either side of the vertebra one on the left and one on the right

Extending from the level of the 12th thoracic vertebra to the 3rd lumbar vertebra.

There is some protection given by the lower ribs.

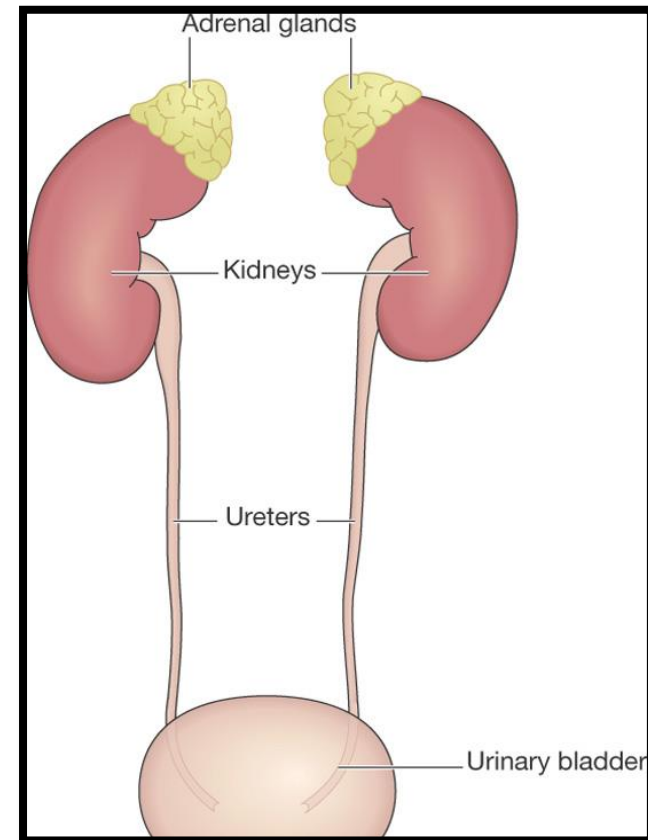


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The Ureters

- The ureters transports urine from the kidneys to the urinary bladder via peristalsis.
- The ureters are between 25cm to 30cm in length.



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The Urinary Bladder

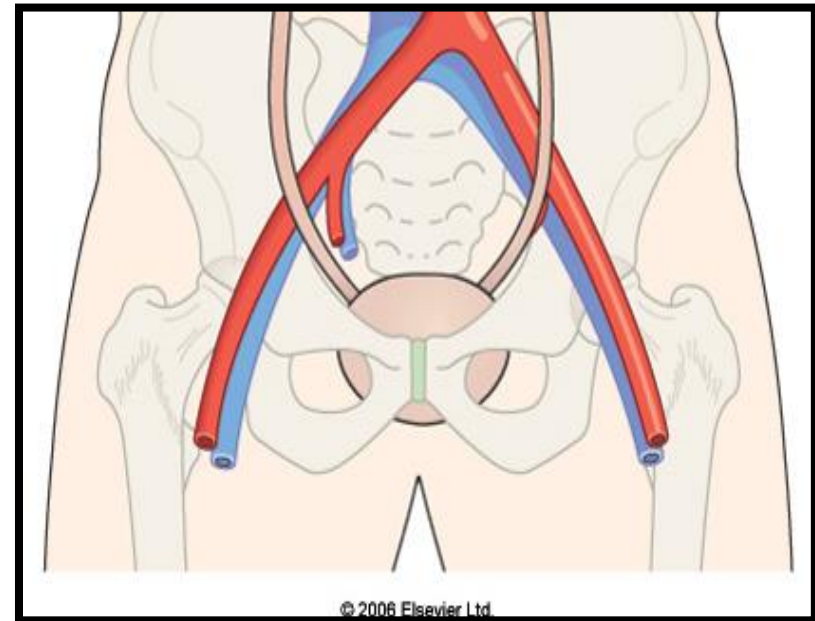
The urinary bladder is the reservoir for urine within the urinary system.

Position

- Within the pelvic cavity, however can rise to the abdominal cavity when full.

Shape and Volume

- Roughly a pear shaped organ, but becomes more balloon shaped when full.
- The total capacity of the bladder is about 600ml.



Urinary System - Part Component 2

The Urethra

- Is a canal that extends from the neck of the bladder to the external urethra orifice.
It is how urine leaves the bladder and is eliminated from the body.
- The male urethra is about 19cm - 20cm long.
- The female urethra is approximately 4cm long and 6mm wide



Urinary System - Part Component 2

Urine

- Urine is clear amber coloured fluid.
- The body excretes 1000 ml to 1500ml of urine a day.
- Urine is composed of:
 - 96% water.
 - 2% Urea.
 - 2% other salts, proteins and hormones.

Urinary System - Part Component 2

Any Questions ?





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Anatomy and Physiology – Part Component 2

Reproductive System



Reproductive System - Part Component 2

Objective

- Explain the functions of the reproductive system
- Know the differences between the male and female reproductive systems

Reproductive System - Part Component 2





Reproductive System - Part Component 2

Functions of the Female Reproductive System

- Formation of the ova
- Reception of spermatozoa
- Provision of suitable environments for fertilisation of foetal development
- Parturition (childbirth)
- Lactation – the production of breast milk

Reproductive System - Part Component 2

External Genitalia - known collectively as the Vulva

Internal Genitalia

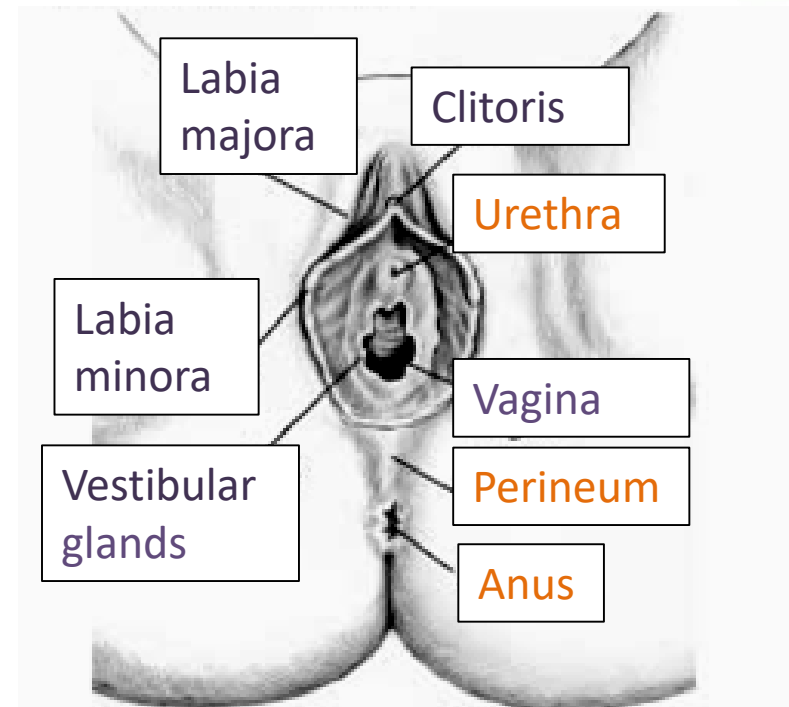


Mammary Glands — accessory glands of the reproductive system

Reproductive System - Part Component 2

External Genitalia:

- Labia majora
- Labia minora
 - Labia majora has two large folds of skin.
 - The labia minora has two smaller folds of skin where the vestibule lies providing secretions.
- Vestibular glands
- Clitoris



Reproductive System - Part Component 2

Internal Genitalia – vagina



- Its function is to receive the penis during sexual intercourse and to allow the delivery of a baby during child birth.
 - Hymen - a thin layer of mucous membrane that partially occludes the vagina – this is torn away by the insertion of a tampon or sexual intercourse.
 - The vagina consists of three layers of tissue: outer - areolar tissue, middle - smooth muscle and inner - squamous epithelium.

Reproductive System - Part Component 2

Internal Genitalia – uterus



- Formed by the fundus, body and cervix.
- **Zygote** (fertilised egg which is formed by the fusion of an ovum and spermatozoon) is embedded and nourished until a placenta is formed.
- It contracts to deliver the baby when in labour.
- The fundus is dome shaped at the top of the uterus and can be felt during pregnancy.

Reproductive System - Part Component 2

Internal Genitalia – uterus



The uterus consists of three layers of tissue:

- outer - perimetrium,
- middle - myometrium
- Inner – endometrium (epithelium supplied with blood vessels).

The endometrium has two layers – functional and basal.

- **Functional** - sheds during menstruation.
- **Basal** - embeds the zygote and provides initial nourishment.

Reproductive System - Part Component 2



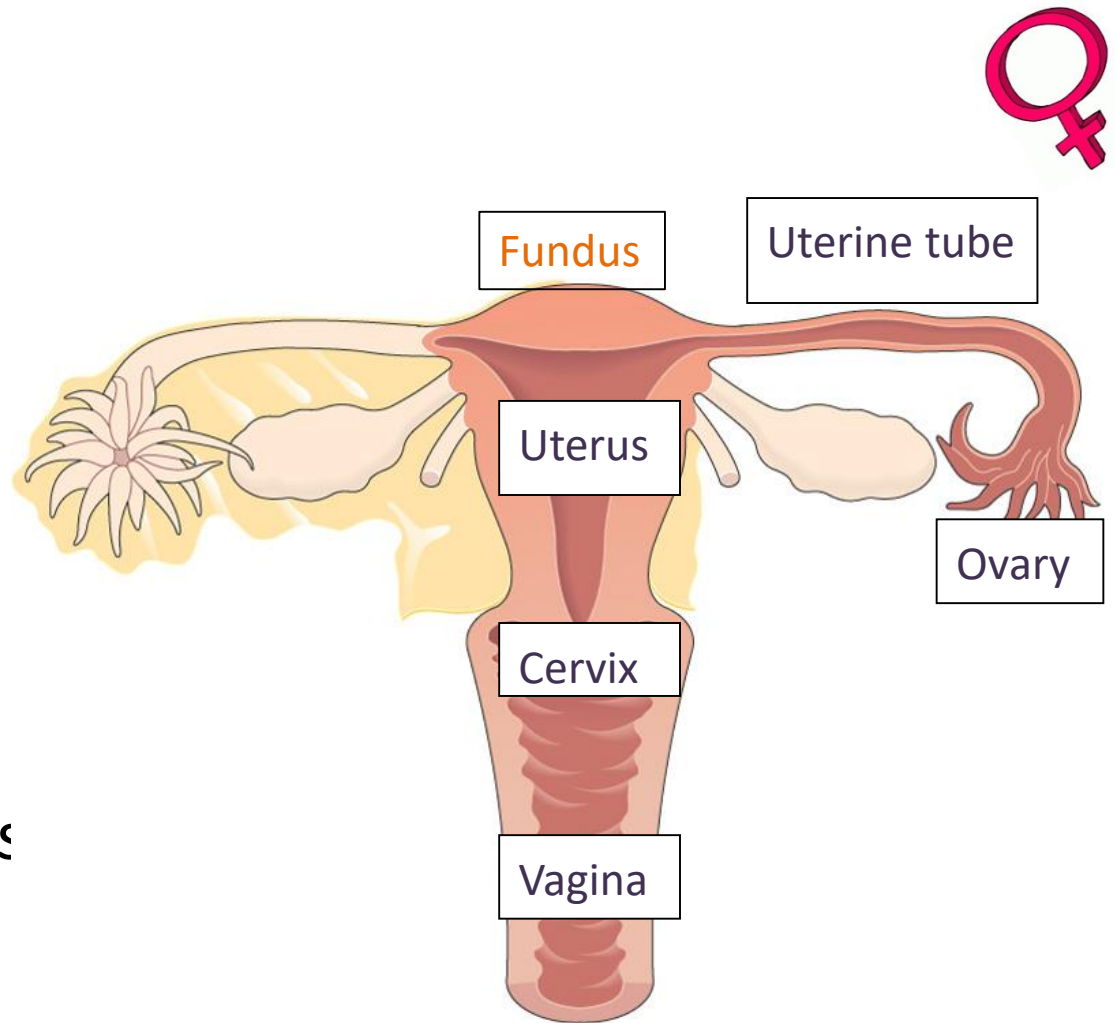
Internal Genitalia – ovaries and uterine tubes

- The ovaries generate sex hormones and the ova (multiple eggs) - no new ova are generated after birth.
- Mature and release the ovum (singular egg) into the uterine tubes
- The uterine tubes move the ovum from the ovary to the uterus by peristalsis and ciliary movement.

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Internal genitalia:

- Vagina
- Cervix
- Uterus
- Uterine tube
- Ovary
- Fundus
- Ligaments, muscles of the pelvic floor



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Resolution: 1280x1024 px
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Reproductive System - Part Component 2

Functions of the Male Reproductive System

- Production, maturation and storage of spermatozoa
- Delivery of spermatozoa in *semen* into the female reproductive tract
- *The urethra is also the passageway for urine excretion*

Reproductive System - Part Component 2

The Penis:



- Two sections:
 - the root (anchors penis in perineum) and the
 - shaft (body of penis visible externally).
- At the tip of the penis is a moveable double layer of skin called the foreskin.
- Formed of erectile tissue and smooth muscle – when stimulated blood flows to the penis and outflow is obstructed, causing an erect penis for intercourse to occur.

Reproductive System - Part Component 2

The Testes:



- Male reproductive glands
- Located in the scrotum outside the body - sperm requires a lower temperature (3 degrees lower) than the rest of the body to generate
- Sperm generation starts at puberty
- Held in place by the spermatic cord
- The deferent duct transports semen from the testes to the seminal vesicle.
- The seminal vesicle duct and deferent duct join to form the ejaculatory duct.

Reproductive System - Part Component 2



The Urethra:

- Pathway for semen and urine to exit the body
- Semen is propelled through the urethra by the ejaculatory ducts and seminal vesicles
- The urethra passes through the prostate gland where milky fluid containing clotting enzymes are added.

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Semen is comprised:

- Seminal fluid 50-60%
- Prostate secretions 20-30%
- Sperm 10%



Reproductive System - Part Component 2

The Prostate:

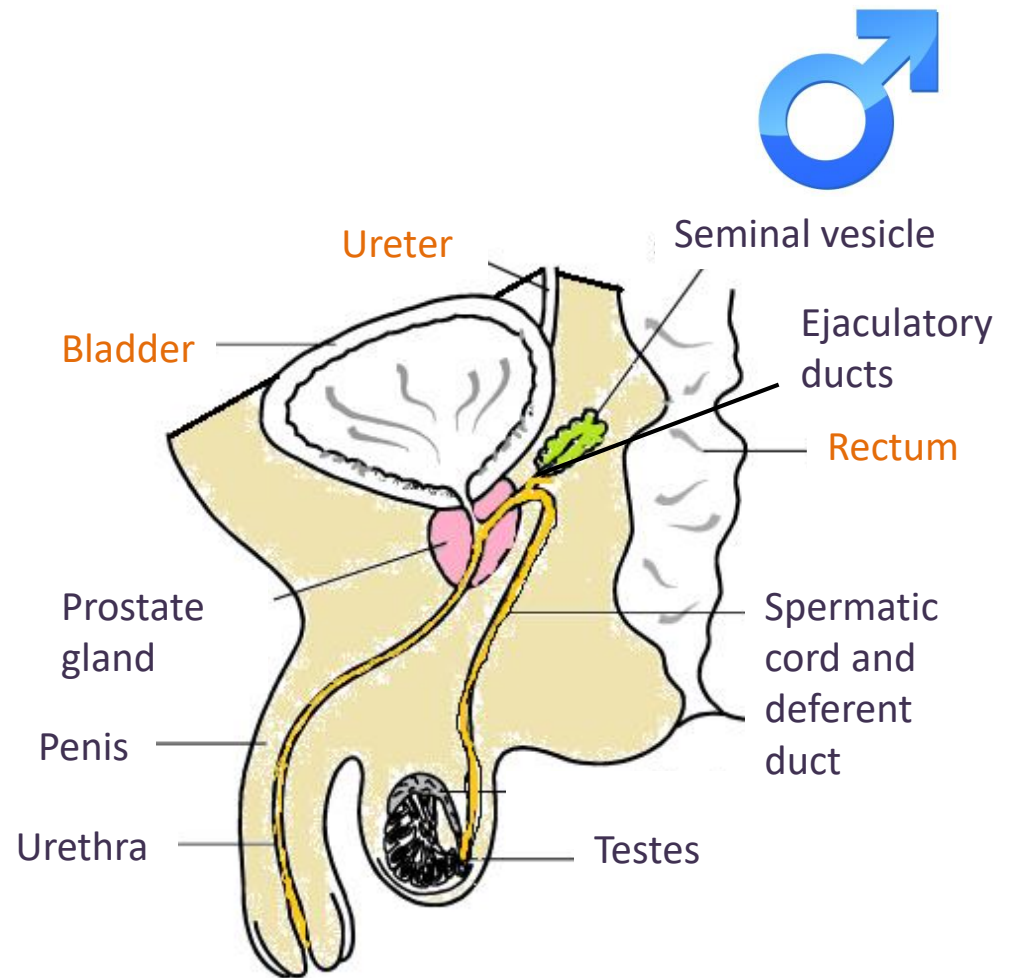


- Located underneath the bladder
- The urethra passes through the prostate
- Consists of a fibrous covering that encloses glandular tissue wrapped in smooth muscle
- Produces a thin, milky fluid containing clotting enzymes to increase likelihood of semen being retained close to the cervix

Reproductive System - Part Component 2

Genitalia:

- Scrotum
- Testes
- Spermatic cords
- Seminal vesicles
- Ejaculatory ducts
- Prostate glands
- Urethra
- Penis



Reproductive System - Part Component 2

Questions?

